

# Day 1 — The Hook: $97 \times 96$ in Three Seconds

**Module:** Nikhilam Multiplication · **Band:** Middle · **Time:** 45 min

## Learning objective

By the end of this lesson, students will record a personal baseline time for  $97 \times 96$  solved by long multiplication, and will have seen the *Nikhilam* method produce the same answer in under 5 seconds.

## Materials

- Printed baseline sheet: a single problem  $97 \times 96 = ?$  with workspace and “your time” line. One per student.
- Stopwatch or wall clock with a second hand. One per pair is enough.
- Whiteboard with a single vertical dividing line drawn down the middle (the “Nikhilam split”).

## Vocabulary introduced today

- *Nikhilam* (IAST: *nikhilam*; lit. “all” — short for *Nikhilam Navataścaramaṁ Daśataḥ*, “all from 9 and the last from 10”) — Tirthaji’s 2nd sūtra; here, the rule for multiplication near a base.
- *deficit* — how much a number falls short of the base (97 is 3 short of 100; deficit = 3).
- *base* — a power of 10 the two numbers are close to (here, 100).

## Lesson flow

### Warm-up (5 min)

- Teacher: “How long does it take you to multiply  $97 \times 96$ ?” Take 3–4 guesses. Write the range (e.g. “15 sec to 1 min”).
- Teacher: “I’ll show you a way to do it in under 5 seconds. Then we’ll talk about why.”

### Baseline timing (10 min)

- Hand out the baseline sheet face-down.
- Students write their name and predicted time on the back.
- On “Go,” flip and solve  $97 \times 96$  using long multiplication.
- Record actual time. Mark answer; honest self-correction. (Correct answer: 9312.)
- Teacher walks around. Most will take 30–60 seconds. A few faster, a few slower.

### Teacher demonstration (10 min)

- On the board, write the problem:

$$\begin{array}{r} 97 \leftarrow (3 \text{ below } 100, \text{ deficit} = 3) \\ \times 96 \leftarrow (4 \text{ below } 100, \text{ deficit} = 4) \\ \hline \end{array}$$

- **Step 1: Find the deficits from 100.** Write  $-3$  next to 97 and  $-4$  next to 96.

$$\begin{array}{r|l} 97 & -3 \\ 96 & -4 \end{array}$$

- **Step 2: Cross-subtract for the LEFT part.** “Take one number minus the *other* number’s deficit.  $97 - 4 = 93$ . (Equivalently:  $96 - 3 = 93$ . Either cross gives the same answer.)”
- **Step 3: Multiply the deficits for the RIGHT part.** “ $3 \times 4 = 12$ .”
- **Step 4: Stick them together.** “93 on the left, 12 on the right  $\rightarrow 9312$ .”

$$\begin{array}{r|l} 97 & -3 \\ 96 & -4 \\ \hline 93 & 12 \end{array} \rightarrow 9312$$

- Verify: long-multiplication gives 9312. ✓
- “*I never multiplied any 2-digit number by another 2-digit number. I only did  $97 - 4$  and  $3 \times 4$ . That’s the entire trick — when both numbers are near a base.*”

### Activity (15 min) — First try

- Hand each student a second sheet with four problems:
  - $98 \times 97$ ,  $96 \times 95$ ,  $99 \times 99$ ,  $97 \times 92$ .
- Students try the Nikhilam method on each, then verify with long multiplication.
- Time themselves. Expect mistakes and slow times on the first try.

### Wrap-up (5 min)

- Show of hands: who got at least one right with Nikhilam? Who was *faster* with Nikhilam already? (Most won’t be — yet.)
- Preview Day 2: “*Tomorrow we’ll work on the foundational skill — instantly finding the deficit of any number from 10 or 100.*”

## Student-facing content

### The Nikhilam Method — Multiplication Near a Base

When both numbers are near a power of 10 (10, 100, 1000):

1. **Find the deficit** of each from the base.
2. **Cross-subtract** for the LEFT part (one number minus the *other*’s deficit).
3. **Multiply the deficits** for the RIGHT part.
4. **Concatenate:** LEFT | RIGHT = the answer.

Today's baseline: my time on  $97 \times 96$  using long multiplication = \_\_\_\_ seconds. My time using Nikhilam (first try) = \_\_\_\_ seconds.

I expect this gap to FLIP by Day 9.

## Homework

- Practice the “complement to 10” reflex: write the deficit of each from 10.
  - $7 \rightarrow$ ,  $8 \rightarrow$ ,  $9 \rightarrow$ ,  $6 \rightarrow$ ,  $5 \rightarrow$ ,  $4 \rightarrow$ ,  $3 \rightarrow$ ,  $2 \rightarrow$ ,  $1 \rightarrow$  \_\_\_\_.
- Do all 9 in under 10 seconds.

## Differentiation

- **Tier up:** Try  $103 \times 104$  (both *above* 100). Note: instead of cross-subtract, you'll need to cross-add. See if you can work out the pattern before Day 4.
- **Tier down:** If 2-digit feels too fast, drop to single-digit near 10 ( $7 \times 8$ ,  $6 \times 9$ ). We'll formally do this on Day 2.

## Teacher notes

- The baseline time is the *single most important* artefact of Day 1. Make sure every student records it honestly. Day 9's retest depends on it.
- Don't promise a specific speedup. Say “you'll likely be faster by Day 9; how much depends on you.”
- Some students will ask “why does this work?” — defer to Day 7. “*Great question. We'll prove it. For now, just notice that it does work.*”
- A few students will try to short-cut and use  $-3 \times -4$  as the right part, getting a sign-error. Wait until Day 5 for the mixed case. For today, all deficits are positive.

## Citation(s) used in this lesson

- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish WA) — Nikhilam multiplication section.
- *Microsoft Word - sutras1.pdf* — sūtra #2: *Nikhilam Navataścaramaṁ Daśataḥ*.